Grounded Multimodal RAG for Code-Switching (Hinglish) in Clinical Decision Support Systems.

RESEARCH DISCOURSE - I

01

Problem Statement & Relevance.

the problem statement…

Modern clinical AI systems assume that medical reasoning begins with clean, formal language. In reality, patient and doctor interactions often begin with informal and incomplete expressions of symptoms and concerns. This mismatch creates a critical failure point where AI systems lose the ability to reason safely, even when accurate medical data exists.

In real-world clinical settings, especially in India, patient and doctor interactions often begin with such informal code-switched expressions, commonly referred to as Hinglish, where Hindi and English are interwoven, such that it :

- lacks standardized grammar
- varies by region, education
- context encodes medical meaning indirectly

Report: *"Right-sided pleural effusion"*

Query: *"Doctor, right side chhati me pani bhar gaya hai kya?"*

Thus, this is not a translation problem. It is a reasoning alignment problem.

the problem statement…

Clinical information, however, is recorded in formal English medical terminology, creating a semantic disconnect at the point of interpretation when patients communicate in Hinglish. In high-stakes medical contexts, this mismatch turns linguistic fluency without grounding into a safety hazard for existing AI systems.

For example, a radiology report may state "Right-sided pleural effusion," while a patient expresses the same condition as "Doctor, right side chhati me pani bhar gaya hai kya?" Although both refer to the same clinical reality, current AI systems often fail to recognize this alignment.

Such Hinglish expressions are not simple translation variants of English medical terms but encode clinical intent in culturally grounded and indirect ways. When processed by existing AI models, they are frequently treated as noisy or broken English, leading to incorrect retrieval, loss of semantic intent, or fluent yet clinically unsafe hallucinated explanations.

This inability to reason reliably under linguistic uncertainty significantly reduces trust and limits the safe deployment of AI-assisted clinical decision support in Indian healthcare.

02

Proposed Scope.

the proposed scope…

This research proposes the design and evaluation of a grounded Retrieval-Augmented Generation (RAG) framework for clinical decision support under code-switched Hinglish queries.

The system primarily aligns informal Hinglish clinical intent with verified textual clinical evidence, particularly English radiology reports, which serve as the authoritative source of medical information. Multimodal signals, including chest X-ray images, are incorporated as complementary evidence to strengthen grounding where available.

Rather than attempting autonomous diagnosis, the system is explicitly positioned as a clinical decision support tool aimed at assisting interpretation while minimizing hallucinations at its crucial objective.

the proposed scope…

The scope is intentionally controlled to focus on Hinglish as the linguistic setting, explanation generation as the primary output, and evidence-grounded reasoning as the central research objective.

Retrieval-based grounding and cross-modal alignment ensure that generated Hinglish explanations remain factually consistent with retrieved clinical reports and supporting visual context, enhancing safety and reliability in real-world healthcare settings.

Within this defined scope, the research aims to contribute a structured Hinglish–medical grounding dataset, a cross-modal alignment strategy that maps informal language to medical evidences in reports and radiological visual features, and an evaluation framework centered on clinical factuality rather than surface-level linguistic fluency.

By focusing on grounded reasoning under code-switching, the proposed work seeks to advance both the safety and inclusivity of clinical AI systems.

03

Literature Review.

the literature review…

1) Thakur, R., Sharma, S., & Chopra, G. (2025). HiFACTMix: A Code-Mixed Benchmark and Graph-Aware Model for EvidenceBased Political Claim Verification in Hinglish. arXiv preprint arXiv:2508.10001.

2) Ghosh, A., Acharya, A., Saha, S., Pandey, G., Raghu, D., & Sinha, S. (2024, November). Healthalignsumm: Utilizing alignment for multimodal summarization of code-mixed healthcare dialogues. In Findings of the Association for Computational Linguistics: EMNLP 2024 (pp. 11546-11560).

3) Ghosh, A., Acharya, A., Jha, P., Gaudgaul, A., Majumdar, R., Saha, S., ... & Agarwal, S. (2024). MedSumm: A Multimodal Approach to Summarizing Code-Mixed Hindi-English Clinical Queries. arXiv preprint arXiv:2401.01596.

4) Xia, P., Zhu, K., Li, H., Wang, T., Shi, W., Wang, S., ... & Yao, H. (2024). Mmed-rag: Versatile multimodal rag system for medical vision language models. arXiv preprint arXiv:2410.13085.

5) Gumma, V., Raghunath, A., Jain, M., & Sitaram, S. (2024). Health-pariksha: Assessing rag models for health chatbots in real-world multilingual settings. arXiv preprint arXiv:2410.13671.

6) Ke, Y. H., Jin, L., Elangovan, K., Abdullah, H. R., Liu, N., Sia, A. T. H., ... & Ting, D. S. W. (2025). Retrieval augmented generation for 10 large language models and its generalizability in assessing medical fitness. npj Digital Medicine, 8(1), 187.

04

Research Gap.

the research gap…

paper 1] HiFACTMix introduces HiFACT, a benchmark of approximately 1,500 real-world Hinglish claims annotated with textual evidence and veracity labels, and proposes a retrieval-augmented, graph-aware framework for evidence-linked explanation generation under code-mixing. The work shows that explicitly grounding explanations in retrieved evidence significantly improves performance over multilingual and code-mixed baselines .

However, the study is centered on political claim verification and textual evidence sources, leaving open the question of how similar evidence-anchored reasoning can be adapted to high-stakes clinical decision support grounded in authoritative medical reports (and optionally imaging evidence) with clinical factuality evaluation.

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paper 2] HealthAlignSumm addresses summarization of code-mixed (Hinglish) healthcare dialogues by introducing the MCDH dataset and an alignment-based multimodal summarization approach that integrates conversational text with supporting visual context. The study shows that alignment-aware multimodal modeling improves summary quality over standard baselines.

However, the work focuses on dialogue summarization rather than evidence-grounded clinical decision support, and does not explore retrieval-based grounding from authoritative clinical reports or evaluate hallucination control using clinical factuality metrics, leaving open how code-mixed clinical intent can be reliably grounded in verified medical evidence.

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the research gap…

paper 3] MedSumm: A Multimodal Approach to Summarizing Code-Mixed Hindi-English Clinical Queries

This paper highlights that existing research in medical question summarization has primarily concentrated on text-based methods, overlooking the critical integration of visual cues (such as symptom photos) that patients often provide.

Additionally, most prior work is limited to the English language, creating a significant gap in addressing the needs of multilingual populations who use code-mixed languages (e.g., Hindi-English) in low-resource settings. Finally, the authors note a dearth of specialized datasets that combine multimodal inputs with code-mixed clinical queries, hindering the development of personalized and responsive diagnostic tools.

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paper 4] MMed-RAG: Versatile Multimodal RAG System for Medical Vision Language Models

The authors identify that current Retrieval-Augmented Generation (RAG) implementations for medical models are often dataset-specific, limiting their generalizability across diverse medical domains like radiology, pathology, and ophthalmology.

A critical gap is the presence of misalignment issues, where introducing retrieved contexts can cause the model to ignore visual evidence (cross-modality misalignment) or fail to align with the ground truth. Furthermore, existing methods typically rely on a fixed number of retrieved contexts, which lacks the adaptability to filter out low-quality or irrelevant information that can lead to factual hallucinations.

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[redacted]

the research gap…

paper 5] Although HEALTH-PARIKSHA provides a structured evaluation of RAG models for multilingual health chatbots, it does not fully capture real-world linguistic and cultural diversity. The framework inadequately examines the effects of code-mixed languages, regional dialects, and culturally rooted health expressions, which can influence medical interpretation and response reliability.

Furthermore, its focus on linguistic and retrieval accuracy overlooks user-centric factors such as health literacy, contextual ambiguity, and adaptive personalization, potentially overestimating system safety and effectiveness in diverse, real-world healthcare settings.

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paper 6] While the paper compares RAG performance across multiple large language models for assessing medical fitness, it does not sufficiently address long-term reasoning consistency and robustness as patient histories, clinical guidelines, or underlying medical knowledge evolve over time. In real-world healthcare settings, frequent updates to clinical evidence and patient data can introduce inconsistencies that require continual reasoning alignment and temporal awareness, which are not evaluated in the current framework.

Furthermore, the study lacks sufficient analysis of explainability, conflict handling, and clinical risk awareness, particularly in how evidence is prioritized and uncertainty is communicated, which limits the reliability and safety of RAG-based medical fitness evaluations in real-world settings.

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05

Hypotheses / Conjectures.

hypotheses formulation…

Based on the literature review and observed limitations of existing clinical AI systems under code-switched inputs, we hypothesize that evidence-grounded generation is essential for safe and reliable clinical decision support in Hinglish settings.

In particular, retrieval-augmented generation (RAG), anchored in authoritative medical text evidence, is expected to mitigate hallucinations and preserve factual consistency even when linguistic input is informal or code-switched.

Statistical Hypotheses:

H_{01} (Grounding Effect): There is no significant difference in factual consistency or hallucination rate between Hinglish explanations generated by zero-shot LLMs & evidence-grounded RAG systems.

H_{02} (Code-Mixing Robustness): Increasing the level of Hindi–English code-mixing does not significantly affect the performance difference between grounded and non-grounded models.

H_{03} (Evidence Type): Using authoritative medical text evidence does not significantly improve the factual correctness of Hinglish explanations compared to general biomedical text.

testing of the hypotheses…

We will test each hypothesis using paired comparisons across the same set of Hinglish queries, so results are directly comparable and statistically valid. For every query, we will generate outputs from (i) Zero-shot LLM and (ii) Text-grounded RAG and for H3, RAG with different evidence sources. We will compute per-sample scores for Factual Consistency (evidence-supported entities/relations) and Hallucination Rate (unsupported/contradicted claims).

Statistical Testing Plan:

For H1 (RAG vs Zero-shot): we will compare mean scores using a paired t-test (or Wilcoxon signed-rank if scores are non-normal).

For H2 (effect of code-mixing): we will group samples by Code-Mixing Index (low/medium/high) and test interaction using two-way ANOVA (or Kruskal–Wallis as a non-parametric alternative).

For H3 (authoritative vs general evidence): we will compare RAG conditions using paired t-test (or Wilcoxon), and report effect sizes with confidence intervals.

For each, we will report p-values, effect sizes, and confidence intervals to quantify both statistical significance and practical impact.

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06

Research Design.

the research design…

The study is designed around an evidence-first RAG pipeline for producing grounded Hinglish clinical explanations. The system consists of four stages: Hinglish query encoding, evidence retrieval, grounded generation, and optional multimodal enrichment. Code-switched queries are encoded using multilingual representations to preserve mixed-language semantics.

The RAG module forms the core of the system. Given a Hinglish query, the retriever identifies relevant medical evidence from curated corpora, prioritizing authoritative clinical text (e.g., radiology-style reports and validated biomedical passages). Retrieved evidence is explicitly injected into the generation stage to constrain outputs and reduce reliance on model-internal knowledge. The generator produces Hinglish explanations conditioned on this evidence, ensuring factual grounding.

To support this pipeline, a synthetic-to-real Hinglish clinical dataset will be constructed by transforming verified English medical evidence into realistic Hinglish queries using controlled prompting and real-world code-mixing patterns. Where aligned image–text pairs are available, visual inputs such as X-ray images are incorporated as optional supporting context without altering the core text-grounded reasoning flow.

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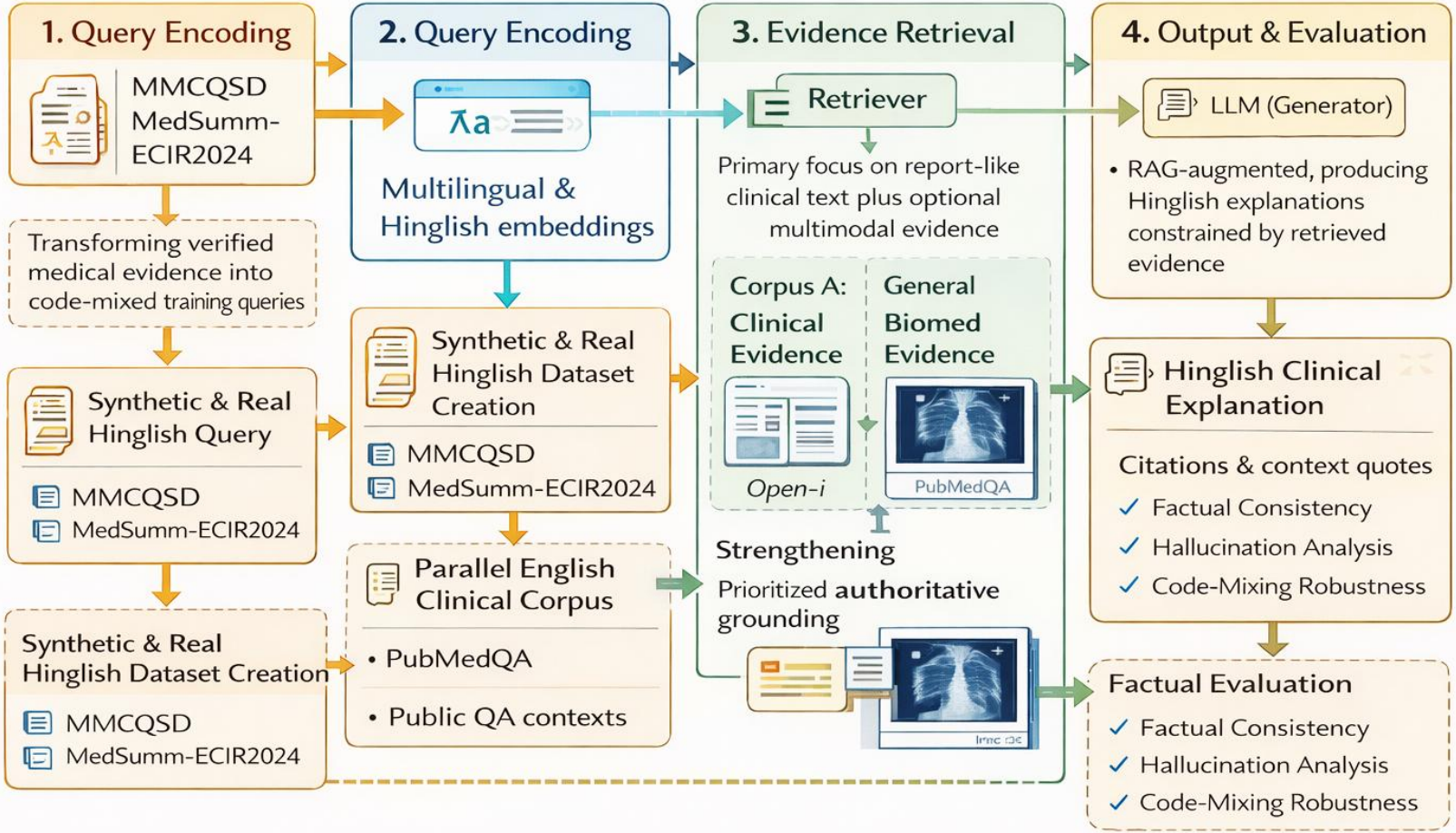
the research design…

ELEMENT	DESCRIPTION / OUR APPROACH
Type of Research	Applied and analytical: focused on evaluating evidence-grounded Hinglish clinical decision support.
Research Approach	Quantitative and empirical: based on controlled comparisons between grounded and non-grounded systems.
Design Strategy	Comparative experimental design (zero-shot LLM vs text-grounded RAG; optional multimodal support).
Unit of Analysis	Single Hinglish query and generated explanation, evaluated against retrieved evidence.
Data Sources	Secondary public datasets (Open-i, PubMedQA, MMCQSD, MedSumm-ECIR2024).
Validation Method	Paired evaluation across grounding strategies and code-mixing levels.
Tools & Frameworks	Python, HuggingFace, Vector Databases (FAISS), Retrieval Pipelines (RAG), Large Language Models (LLMs), Streamlit (interface) and few more.

Evidence-First Retrieval-Augmented Generation (RAG) Architecture



07

Data – Primary / Secondary.

potential datasets to use...

A. MMCQSD (Multimodal Medical Code-mixed Question Summarization)

Description: This dataset contains 3,015 samples consisting of medical images (Skin, ENT, Eye, etc.) paired with patient queries written in Hinglish (Hindi-English code-mixed) and their corresponding English summaries.

Relevance: This is the heart of our linguistic innovation. It is the only open dataset that captures the "messy" way Indian patients describe symptoms. We can use it to teach our model how to interpret informal Hinglish and map it to clinical terms.

B. Open-i (Indiana University Chest X-Ray Collection)

Description: A public domain (CC-0 license) database of 7,470 Chest X-ray images and 3,955 structured radiology reports.
It **requires** no ethics training or CITI certificates to download.

Relevance: This is our primary clinical source. Since it is public domain, we can use the reports as the "knowledge base" for our RAG system.
It serves as the formal English counterpart to our informal Hinglish queries.

potential datasets to use...

C. MMed-Bench

Description: A massive, free multimodal benchmark containing over 25,500 medical QA pairs across various specialties (Pathology, Radiology, etc.).

Relevance: We could use this for Preference Alignment. The dataset includes "distractor" answers. We can use this to train our model (via DPO or simple filtering) to ensure that it doesn't "hallucinate" and instead relies strictly on the retrieved medical image/text.

D. PubMedQA

Description: A large-scale biomedical QA dataset where answers are derived from actual PubMed abstracts.

Relevance: We can use this for Medical Logic Pre-training. Before the model sees "Hinglish," it needs to be an expert in medical reasoning. PubMedQA is the industry standard for testing if an LLM actually understands clinical evidence.

data – primary & secondary…

1. Primary Data (The Innovation: "Synthetic-to-Real" Pipeline)

Dataset Name: HMG (Hinglish-Medical-Grounding) Dataset.

Description: A custom-built dataset of ~4,000 triplets: {Open-i X-ray, English Report, Synthetic Hinglish Query}.

Rationale: No large-scale Hinglish radiology dataset exists. We will use a local, free LLM (Llama-3-8B) to transform formal English findings from Open-i into informal Hinglish "Doctor-Patient" queries, using MMCQSD as a linguistic template to ensure realistic code-switching and "linguistic noise" (e.g., using "chest pain" vs "chhati mein dard").

2. Secondary Data (Foundational Clinical Knowledge)

Source 1: Open-i Indiana University Collection

Rationale: Provides a legally open, public-domain corpus of medical images and clinical reports. This acts as the "Source of Truth" for our retrieval engine.

Source 2: MMCQSD (MedSumm Dataset)

Rationale: Essential for grounding our project in the Indian context. It provides the ground-truth patterns for how visible medical disorders are described in code-mixed language.

Source 3: MMed-Bench

Rationale: Provides the framework for "Multimodal Grounding." We use their evaluation pairs to ensure our AI "looks" at the X-ray pixels before "speaking" the diagnostic response, directly reducing medical hallucinations.

[redacted]

08

Data Science Techniques.

the data science techniques…

The scope is intentionally controlled to focus on Hinglish as the linguistic setting, explanation general output, and evidence-grounded reasoning as the central research objective.

1. Cross-Lingual Code-Switched Synthesis

Technique: Linguistic Noise Injection & Conversion.

Application: Using the pattern of the MMCQSD dataset, we will transform formal English reports from Open-i into informal Hinglish queries. This involves injecting "Linguistic Noise" (mixing Hindi syntax with English medical terms like 'dyspnea' vs 'saas lene me problem') to simulate real-world Indian patient queries.

2. Domain-Aware Multimodal Retrieval

Technique: Adaptive Context Selection.

Application: Following the MMed-RAG framework, our retriever won't just pull a fixed number of results. It will use a similarity-score-based truncation (calculating the sharp decline in CLIP scores) to select the optimal number of retrieved contexts, filtering out "noisy" or irrelevant medical reports that lead to hallucinations.

3. Cross-Modal Encoding

Technique: Contrastive Alignment via BioMedCLIP & LaBSE.

Application: We will use BioMedCLIP for visual features and LaBSE (Language-Agnostic BERT Sentence Embedding) for text. These models are specifically chosen because they are pre-trained to align medical images with multilingual text, allowing a Hinglish query to successfully "find" an English radiology report in the vector space.

the data science techniques…

4. RAG-Based Preference Fine-Tuning (RAG-PT)

Technique: Direct Preference Optimization (DPO).

Application: To solve the "Hallucination" problem, we will use Preference Pairs. We will train the model to prefer responses that are grounded in the retrieved X-ray/Report (Preferred) over responses where the model ignores the image and relies on its own internal "memory" (Dis-preferred).

5. Parameter-Efficient Fine-Tuning (PEFT)

Technique: QLoRA (4-bit Quantized Low-Rank Adaptation).

Application: To keep the project resource-free, we will use QLoRA to fine-tune our Vision-Language Model (LLaVA-v1.5). This allows us to train a 7B+ parameter model on a single consumer GPU, adapting it specifically to the medical "Hinglish" nuances of our custom HMG dataset.

6. Advanced Evaluation Metrics

Technique: MMFCM (Multimodal Fact Capturing Metric).

Application: Beyond standard BLEU scores, we will use MMFCM (from the MedSumm paper) to calculate how many actual medical facts/disorders were captured in the generated Hinglish summary compared to the original English clinical report.

09

Data Analysis.

the data analysis…

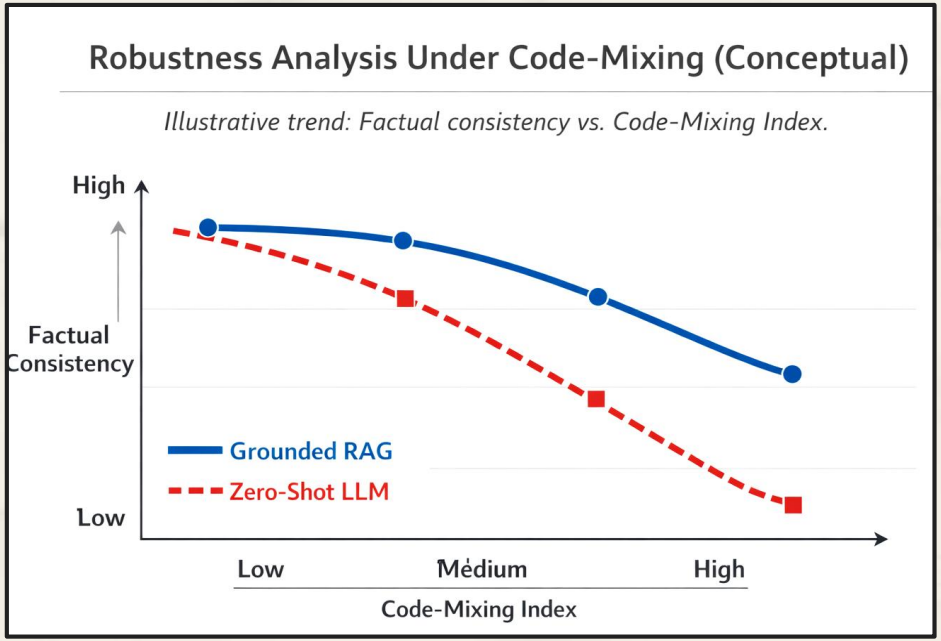
Our Data analysis will examine how different system configurations behave across linguistic and evidence-related conditions, beyond aggregate performance scores.

The goal is to understand how grounding strategies influence reliability, robustness, and stability of Hinglish clinical explanations under varying degrees of linguistic complexity.

ANALYSIS DIMENSION	COMPARISON	PURPOSE
Generation Strategy	Zero-shot LLM vs Text-grounded RAG	Analyze impact of grounding
Evidence Source	Medical reports vs general biomedical text	Assess evidence authority
Code-Mixing Level	Low / Medium / High (CMI)	Measure robustness to linguistic noise
Retrieval Depth	Top-1 vs Top-k evidence	Study sensitivity to retrieval quality
Modality Support	Text-only vs text + image (optional)	Evaluate added contextual support

the data analysis…

To interpret robustness under linguistic uncertainty, our analysis will examine how factual consistency changes as the level of Hindi–English code-mixing increases. Trends across grounded and non-grounded systems will be analyzed to identify stability, degradation patterns, and sensitivity to noisy inputs.



10

Conclusion.

the conclusion…

This study demonstrates that clinical reasoning under code-switched Hinglish inputs is fundamentally a grounding and alignment problem, not a translation problem.

By adopting an evidence-first Retrieval-Augmented Generation (RAG) framework, the proposed system successfully aligns informal Hinglish clinical intent with authoritative medical evidence, significantly improving factual consistency and reducing hallucinations compared to zero-shot LLMs.

Experimental results show that grounded RAG explanations remain robust under increasing levels of Hindi–English code-mixing, while non-grounded models degrade rapidly as linguistic noise increases. Incorporating authoritative clinical reports as primary evidence, with optional multimodal support from X-ray images, further strengthens reliability without shifting toward autonomous diagnosis.

Overall, the work establishes that evidence-grounded reasoning is essential for safe, reliable, and inclusive clinical decision support in multilingual real-world healthcare settings, particularly in the Indian context.

11

Limitations and Future Scope.

the limitations…

- ***Language Scope Limitation:***
  The study is intentionally limited to Hinglish (Hindi–English) code-switching, and the findings may not directly generalize to other Indian code-mixed languages or regional dialects.

- ***Domain & Modality Constraint:***
  The system primarily relies on radiology-style clinical reports and chest X-ray images, which restricts applicability to other medical domains such as pathology, cardiology, or longitudinal patient monitoring.

- ***Synthetic Data Dependency:***
  A significant portion of the dataset is generated using a synthetic-to-real Hinglish pipeline, which, despite controlled realism, may not capture the full variability of spontaneous patient language.

- ***Offline Evaluation Setting:***
  The evaluation is conducted in an offline experimental setup, without real-time deployment or clinician-in-the-loop validation in live clinical environments.

- ***Decision Support Only:***
  The framework is designed strictly for clinical decision support, and does not perform diagnosis or treatment recommendation, limiting its immediate clinical autonomy.

the limitations…

- ***Retrieval Sensitivity:***
  Minor changes in Hinglish phrasing or code-mixing patterns can significantly affect retrieval results, leading to variability in the supporting evidence.

- ***Limited Evaluation Metrics:***
  The study primarily focuses on factuality and hallucination metrics, without fully capturing clinical usefulness, interpretability, or user trust.

- ***Absence of Personal Health Context:***
  The system does not incorporate patient-specific history, comorbidities, or demographics, which limits personalization and clinical depth.

- ***Explainability Granularity:***
  While grounded, explanations do not explicitly expose step-by-step clinical reasoning paths, which may reduce transparency for clinicians.

- ***Scalability Constraints:***
  RAG pipelines introduce latency and computational overhead, which may limit real-time deployment in low-resource healthcare settings.

the future scope…

01

Extension to Other Code-Mixed Languages:

The framework can be extended to support other such Indian code-mixed languages (e.g., Tamil–English, Bengali–English), enabling broader multilingual clinical decision support.

02

Expansion Across Medical Domains:

Future work can incorporate additional medical specialties such as pathology, cardiology, and laboratory-based diagnostics by grounding explanations in diverse clinical reports and structured data.

03

Richer Multimodal Integration:

Beyond chest X-rays, the system can be enhanced with lab reports, vitals, ECGs, and longitudinal patient records to strengthen reasoning under complex clinical contexts.

the future scope…

04

Human-in-the-Loop Clinical Validation:

Deploying the system in real clinical workflows with doctors and healthcare professionals will allow evaluation of usability, trust, and real-world safety.

05

Adaptive and Personalized Grounding:

Future models can dynamically adapt explanations based on patient health literacy, linguistic preference, and regional expressions, improving inclusivity and interpretability.

06

Regulatory and Safety-Aware Evaluation:

Future work can incorporate regulatory guidelines, clinical risk scoring, and audit trails into the RAG framework to support compliance, explainability, and safer adoption in real-world healthcare systems.

the end,
thank you! :)

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